Supplementary Material for "Towards Accurate and Lightweight Peripheral Neuroblastic Tumor Diagnosis via Contrastive Multi-scale Pathological Image Analysis"

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I. Additional BreakHis Analysis at 40× Magnification

To further evaluate cross-magnification generalization, CoHisNet was assessed on the public BreakHis dataset [1] using the 40× magnification setting. This experiment provides a lower-magnification complement to the 400× BreakHis analysis in the main manuscript and is closer to the 10×/20× scanning context of the private PpNTs cohort, although BreakHis differs from PpNTs in tumor type, category definition, and imaging protocol. As shown in Table I, CoHisNet-tiny achieved the best performance across all five metrics, with 50.83% ACC, 45.51% BACC, 44.96% Kappa, 45.09% F1, and 83.84% AUROC. Among the non-CoHisNet baselines, CONCH achieved the second-best ACC, whereas KEEP achieved the second-best BACC, Kappa, F1, and AUROC. These results indicate that the compact cross-scale design of CoHisNet remains competitive under a lower-magnification public-dataset setting, while the reduced absolute scores also emphasize that BreakHis 40× is a challenging out-of-domain stress test rather than a direct surrogate for pNT diagnosis.

II. WSI-Level Inference Time Analysis

Practical WSI-level efficiency was measured on the internal test set. The complete online pipeline included image loading and preprocessing, CoHisNet patch inference, tissue-component prediction using the trained Prior Synergy Block, pathology-prior weight calculation, and final WSI-level voting. All measurements were performed under identical experimental settings using an Intel Xeon Gold 6326 CPU and an NVIDIA Tesla V100-SXM2 32GB GPU, with one-time model initialization and loading excluded from the reported runtime.

For a WSI containing N patches and C diagnostic classes, the final weighted aggregation has a computational complexity of $\mathcal{O}(NC)$. When patch inference and tissue-component prediction are included, the complete pipeline can be expressed as $\mathcal{O}[N\,(F_{\mathrm{CoHisNet}} + F_{\mathrm{prior}} + C)]$, where F_{CoHisNet} and F_{prior} denote the per-patch computational costs of CoHisNet and the Prior Synergy Block, respectively. As shown in Table II, the final voting arithmetic required less than 1 ms per WSI in all settings. With eight CPU workers, PathVote processed each WSI in 11.439 s on average, corresponding to 120.0 patches/s, while hard voting required 4.506 s per WSI under the same loading and preprocessing setting. The additional runtime and memory of PathVote mainly arose from tissue-component prediction and associated feature processing rather than the final aggregation operation.

III. Analysis of Misclassification Patterns

For error analysis, we examined cases that were consistently misclassified across multiple architectures. Although the ground-truth diagnoses for these cases were poorly differentiated neuroblastoma, baseline models frequently predicted them erroneously as differentiating neuroblastoma. Expert pathological review of these discordant cases identified three recurrent confounding factors underlying such misclassifications:

- **a) Morphologic mimicry by large anaplastic cells.** Misclassified cases frequently contained large multinucleated anaplastic cells, a subset of which exhibited abundant cytoplasm. These elements closely resemble the cytologic features of differentiating neuroblasts, thereby introducing visual ambiguity that biases baseline models toward a differentiating classification.
- **b) Therapy-induced cytoplasmic expansion.** A substantial proportion of tumor cells in misclassified cases displayed voluminous cytoplasm consistent with post-treatment morphologic alteration. Such changes are highly variable across patients and are not universally present in treated neuroblastomas. In these instances, treatment-related modifications introduced morphologic

This work was supported by the National Key R&D Program of China under Grant 2023YFC2706400, the Fundamental Research Funds for the Provincial Universities of Zhejiang under Grant GK259909299001-006 and Anhui Provincial Joint Construction Key Laboratory of Intelligent Education Equipment and Technology under Grant IEET202401. *(Zhu Zhu, Shuo Jiang, and Jingyuan Zheng contributed equally to this work.) (Corresponding authors: Feiwei Qin, Jinhu Wang and Gang Yu.)*

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TABLE I

COMPARISON OF COHISNET WITH REPRESENTATIVE GENERAL-PURPOSE CLASSIFIERS, PATHOLOGY FOUNDATION MODELS, AND PATHOLOGY-SPECIFIC CLASSIFICATION MODELS ON THE BREAKHIS DATASET AT 40× MAGNIFICATION. THE BEST AND SECOND-BEST RESULTS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY.

Model	ACC(%)	BACC(%)	Kappa(%)	F1(%)	AUROC(%)
General method					
VGGNet [2]	47.74	20.29	2.79	14.22	68.42
ResNet-50 [3]	44.51	20.90	10.30	13.30	60.91
DenseNet [4]	39.48	22.95	1.51	14.24	68.75
Vision Transformer [5]	44.64	21.46	3.11	13.20	63.51
Swin Transformer [6]	41.41	23.08	7.77	21.21	70.87
Pathological large-scale pre-trained model					
UNI [7]	42.19	21.13	12.51	22.96	57.42
UNI2 [7]	32.25	29.17	17.68	26.18	60.09
CTransPath [8]	32.64	28.01	9.60	22.87	70.24
CONCH [9]	<u>48.51</u>	28.51	27.10	28.08	79.17
KEEP [10]	46.97	<u>42.09</u>	<u>30.36</u>	<u>37.35</u>	<u>80.57</u>
Pathological classification model					
HisImage [11]	42.83	27.35	26.64	26.19	69.22
FMDNN [12]	45.67	35.17	28.85	33.79	72.79
StoHisNet [13]	46.70	37.43	22.08	34.73	79.85
HiFuse [14]	48.25	25.88	26.47	21.75	76.51
Ours					
CoHisNet-tiny	**50.83**	**45.51**	**44.96**	**45.09**	**83.84**

TABLE II

WSI-LEVEL ONLINE INFERENCE EFFICIENCY UNDER IDENTICAL DATA AND HARDWARE SETTINGS.

Method	CPU workers	Mean (s/WSI)	Median (s/WSI)	Throughput	GPU peak (MiB)	CPU RSS (MiB)	Vote (ms/WSI)
Hard vote	1	9.449	9.246	145.3	572	1,202.0	0.347
PathVote	1	11.953	12.512	114.9	3,948	4,586.3	0.524
Hard vote	8	4.506	4.662	304.7	572	1,284.4	0.477
PathVote	8	11.439	10.650	120.0	3,948	3,849.5	0.640

Note: Throughput is reported as patches/s. The single-worker and eight-worker settings refer to CPU image loading and preprocessing; GPU inference used the same V100 in all experiments.

features overlapping with those of differentiating neuroblastoma, leading to diagnostic confusion in baseline models.

- **c) Non-neoplastic tissue confounders.** Attention map analysis of misclassified cases revealed that regions contributing most strongly to erroneous predictions typically corresponded to normal adrenal cortex or pericapsular inflammatory infiltrates rather than neoplastic tissue. These non-tumor areas were often adjacent to abundant Schwannian stroma, collectively forming a histologic context that mimics the stroma-rich architecture characteristic of differentiating neuroblastoma. Notably, at high magnification, adrenal cortical cells exhibit nuclear and cytoplasmic features overlapping with those of neuroblastic tumor cells, rendering them particularly susceptible to misidentification by convolutional feature extractors.

Fig. 1 illustrates the canonical histopathological phenotype of differentiating neuroblastoma alongside the representative visual patterns corresponding to each of the aforementioned confounding factors (a–c). Collectively, these findings demonstrate how rare morphologic variants, treatment-related alterations, and anatomically adjacent normal tissues can converge to produce systematic misclassification in baseline models. Notably, our proposed PathVote correctly classified these challenging cases as poorly differentiated neuroblastoma, demonstrating superior robustness over conventional approaches in handling morphologically ambiguous and confounding histologic contexts.

To complement the ROC-based evaluation under class imbalance, Fig. 2 presents the class-wise precision–recall curves for the PpNTs internal test set and the BreakHis 400× test set. The upper panel shows the results on the PpNTs internal test set, while the lower panel reports those on BreakHis at 400× magnification. These curves characterize the precision–recall trade-off across decision thresholds and provide additional class-specific information, particularly for less prevalent categories. Together with BACC, macro-F1, Kappa, and AUROC, this analysis provides a more comprehensive assessment of classification performance under class imbalance.

IV. GOLD STANDARD PATHOLOGICAL DIAGNOSIS FOR PNTS

The diagnostic gold standard for peripheral neuroblastic tumors (pNTs) in this study follows the International Neuroblastoma Pathology Classification (INPC) system, an internationally validated framework based on the Shimada classification [15] with established prognostic significance. Histopathological diagnosis was performed via light microscopy accord-

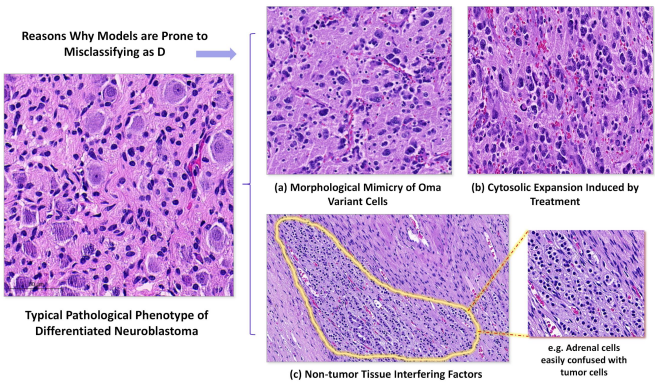


Fig. 1. Representative histopathological phenotypes underlying misclassification in baseline models.

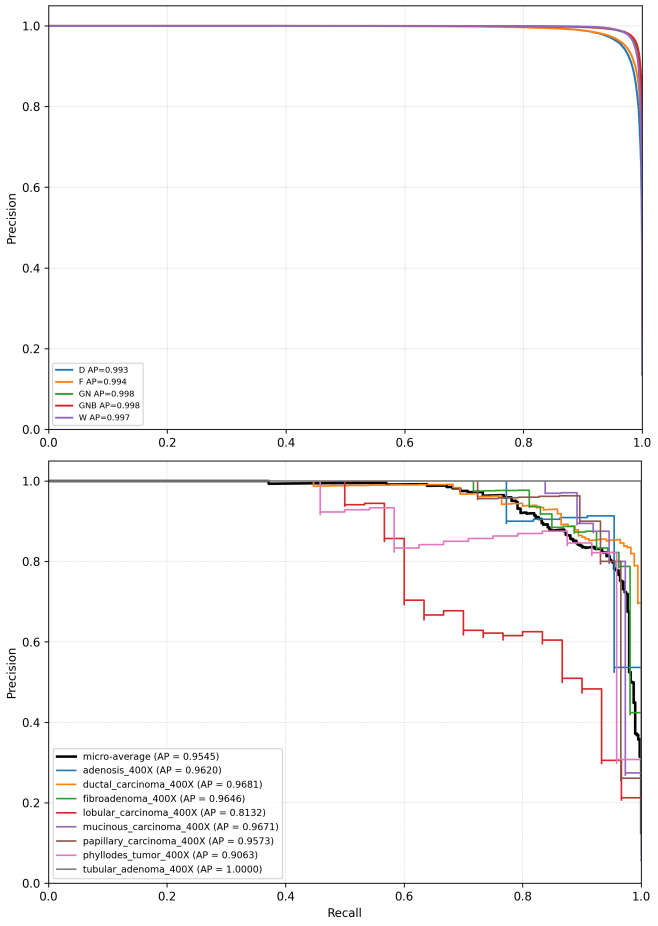


Fig. 2. Precision-recall curves for the five-class PpNTs internal test set (top) and the eight-class BreakHis dataset at 400× magnification (bottom).

ing to a two-tiered workflow: initial categorization into major tumor types followed by histological subtyping for malignant entities. The complete criteria are summarized in Tables III and IV, with key diagnostic logic elaborated below.

The primary diagnostic step is the assessment of neuropil and nested cellular architecture, which serves as the fundamental discriminator between benign and malignant pNTs (Table III). Tumors lacking both neuropil and nested architecture are classified as ganglioneuroma (GN), a benign entity characterized by fully mature ganglion cells, abundant Schwannian stroma, and frequent lymphocytic infiltration. In contrast, the presence of neuropil confirms malignancy and necessitates further quantification of tumor components relative to Schwannian stroma to distinguish neuroblastoma (NB) from ganglioneuroblastoma (GNB). Specifically, NB is diagnosed when tumor components constitute $\geq 50\%$ of the tissue with $< 50\%$ Schwannian stroma, whereas GNB is defined by the inverse proportion ($< 50\%$ tumor components and $\geq 50\%$ Schwannian stroma). Among GNB subtypes, nodular GNB (GNBn) requires gross specimen identification of distinct neuroblastomatous nodules. Due to the unavailability of such morphological data in our retrospective cohort, GNBn cases were excluded to ensure label reliability. Only intermixed GNB (GNBi) and other eligible subtypes were retained.

For tumors classified as NB, histological subtyping is determined exclusively by the proportion of differentiating neuroblasts, which directly correlates with biological aggressiveness (Table IV). Differentiating neuroblasts are operationally defined as cells exhibiting clear cytoplasmic eosinophilia and nuclear maturation distinct from primitive neuroblasts. Undifferentiated NB (UD) shows no evidence of neuronal differentiation and represents the most aggressive subtype. Poorly differentiated NB (PD) contains $\leq 5\%$ differentiating neuroblasts with minimal maturation features, while differentiated NB (D) contains $> 5\%$ differentiating neuroblasts with discernible nucleoli and abundant eosinophilic cytoplasm, corresponding to the lowest malignancy grade among NB subtypes. This morphology-based classification system provides the ground-truth labels for all pNTs included in our dataset, ensuring consistent and clinically relevant annotations for model training and validation.

V. Detailed Age-Stratified Patient Profile

As shown in Fig. 3, this sunburst chart visualizes the granular demographic and pathological breakdown of the study cohort by single-year age intervals, providing a higher resolution view than the aggregated groups presented in the main text. The inner ring represents broader age clusters, while the outer ring details specific annual increments from birth to 18 years. Each sector displays the number of patients corresponding to each age group from birth to 18 years, with the 15–16 and 16–17 age groups missing due to lack of patients in those ranges. It can be observed that cases are significantly concentrated in the 3- to 6-year-old age group, which aligns with the typical incidence peak for neuroblastoma. Color coding indicates the histological subtype, revealing distinct patterns such as the predominance of differentiated tumors in older children and the presence of undifferentiated subtypes primarily in infants and toddlers.

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TABLE III
INPC-BASED DIAGNOSTIC CRITERIA FOR MAJOR CATEGORIES OF PNTS.

Category	Neuropil / Nested Arch.	Tumor (%)	Stroma (%)	Key Histological Features
GN	Absent	N/A	Predom.	Mature ganglion cells; no nests; abundant Schwannian stroma; frequent lymphocytic infiltration. Benign.
NB	Present	$\geq$50	$<$50	Malignant; further subtyped by neuroblastic differentiation degree (see Table IV).
GNBi	Present	$<$50	$\geq$50	Malignant tumor components intermixed with abundant Schwannian stroma.
GNBn	Present	Variable	Variable	Distinct NB nodules within GN/GNBi background. Excluded from this study.

* GNBn exclusion is due to unavailability of gross morphological data required for nodule confirmation.

TABLE IV
HISTOLOGICAL SUBTYPING CRITERIA FOR NB.

Subtype	Diff. Neuroblasts (%)	Malignancy Grade	Key Morphological Notes
UD	0	Highest	No evidence of neuronal differentiation; most aggressive subtype.
PD	$\leq$5	Intermediate	Minimal neuronal differentiation; scant cytoplasmic eosinophilia.
D	$>$5	Lowest	Clear neuronal differentiation; abundant eosinophilic cytoplasm; discernible nucleoli.

Note: Malignancy decreases sequentially from UD to D. Differentiating neuroblasts are defined as cells showing clear cytoplasmic eosinophilia and nuclear maturation distinct from primitive neuroblasts.

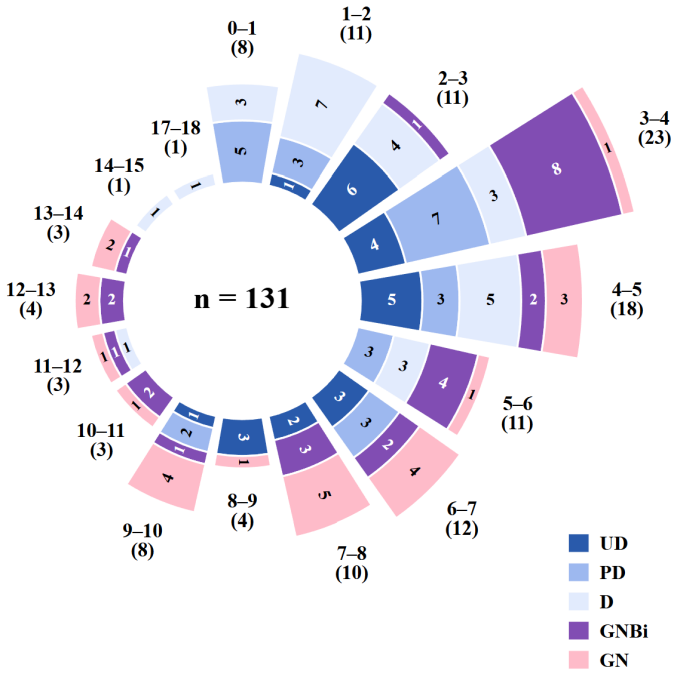


Fig. 3. Detailed age-stratified distribution of clinicopathological characteristics in the neuroblastoma cohort (n=131).

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